

BIOL 1414 Intro to Mol/Cell Biology

Lecture #6

9/10 September 2026

Last Lecture:

- Biological Macromolecules

Today: (Ch 4)

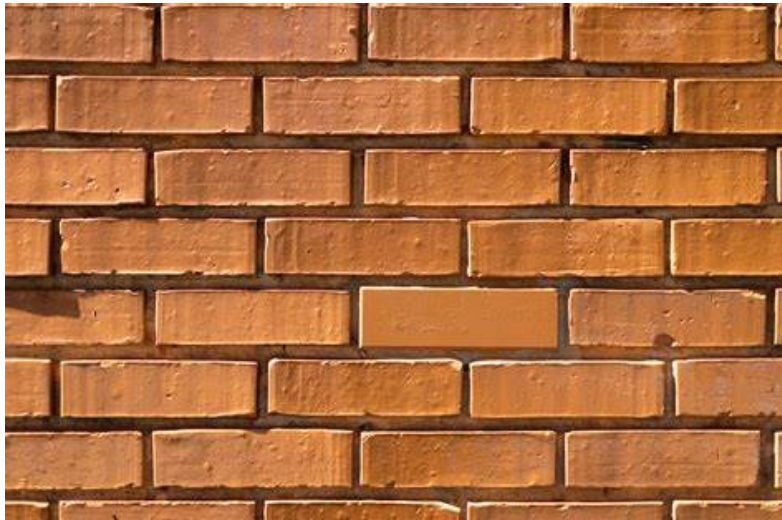
- Introduction
- Studying Cells
- Cell Theory
- Prokaryotes
- Eukaryotes
 - Plant vs. Animal Cells
- Endomembrane system
- Exoskeleton
- Extra cellular Matrix



Myanmar snub-nosed monkey
Rhinopithecus strykeri

Introduction

- Take a second close your eyes and think about a brick wall.
 - What is the basic component that makes up the wall?
 - The brick
 - What is the basic building block of our bodies?
 - The cell

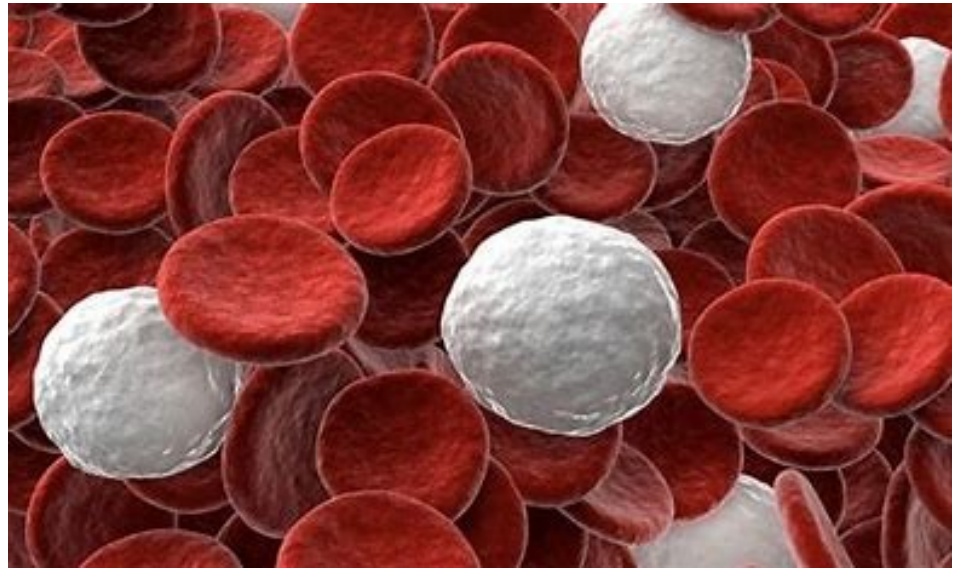


Introduction

- Body consists of many types of cells
 - Each is specialized for a specific purpose
- Compare to building a house
 - Need a wide variety of different materials
 - Each has a specific purpose as well
- Examples of cells:
 - Epithelial – protect body surface and cover organs
 - Bone – help support and protect the body
 - Blood – carry nutrients and gas through body
- Cells from all organisms share fundamental characteristics between them

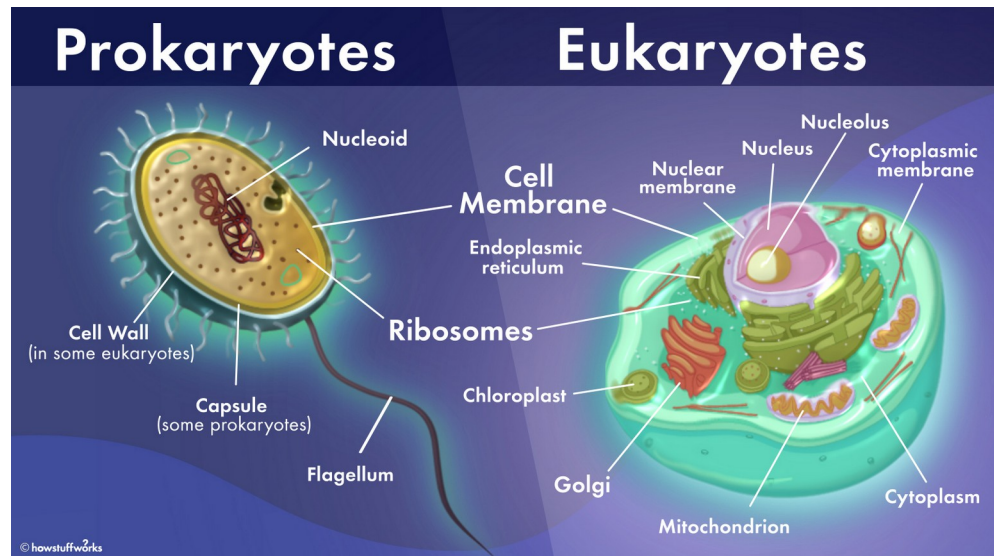
Introduction

- Cells function in:
 - Growth of an organism
 - Development of an organism
 - Day-today maintenance
 - Reproduction
 - Much more



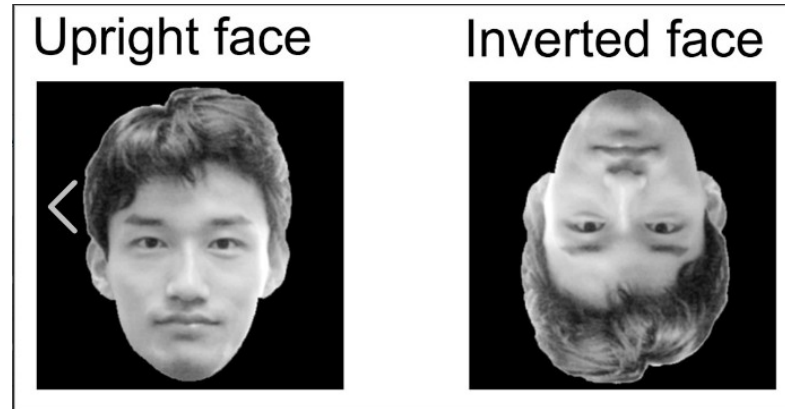
Studying Cells

- Cell is the smallest unit of a living thing
 - Basic building blocks of all organisms
 - Single celled bacteria
 - Complex multicellular human
- Cells classified into two groups
 - Prokaryotes
 - Eukaryotes



Studying Cells

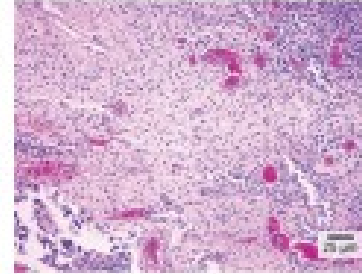
- Microscope:
 - Micro – small
 - Scope – to look at



- Image is inverted in a microscope
 - Right side up and facing right to naked eye
 - Will be upside down and facing left through scope
 - Due to compound scope
 - Two sets of lenses to magnify image
 - As light travels through these lenses it is inverted

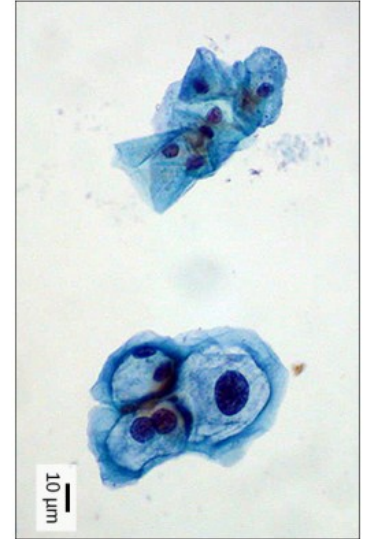
Studying Cells

- Light microscope
 - What we use in lab
 - Many advantages
 - 40x to 1000x magnification
 - Light aids in viewing cells as they are trans
 - Small and easy to work with
 - Disadvantages
 - Usually need to stain cells to see organelles
 - Have lower resolving power
- Electron microscopes
 - What scientist use to study cells
 - More magnification and resolution
 - More expensive and very large



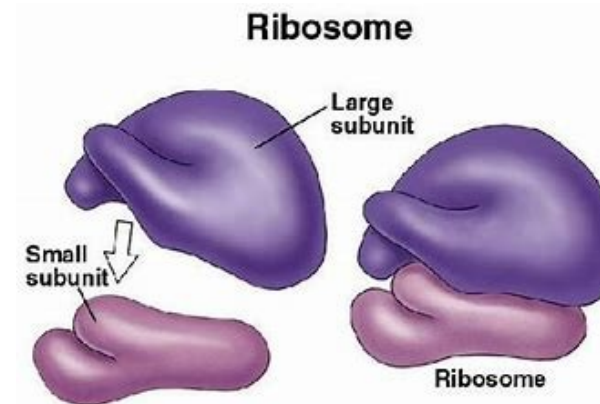
Cell Theory

- **Unified Cell Theory:**
 - All living things are comprised of cells
 - One or more cells
 - The cell is the basic unit of life
 - New cells arise from existing cells
- **Career connection:**
 - Pap smear – test of the uterine cervical cells
 - Cytotechnologists:
 - Scientist who study cells via microscopes
 - » Cyto = cell
 - Determine if cell changes are normal or not
 - Test more than just cervical cells



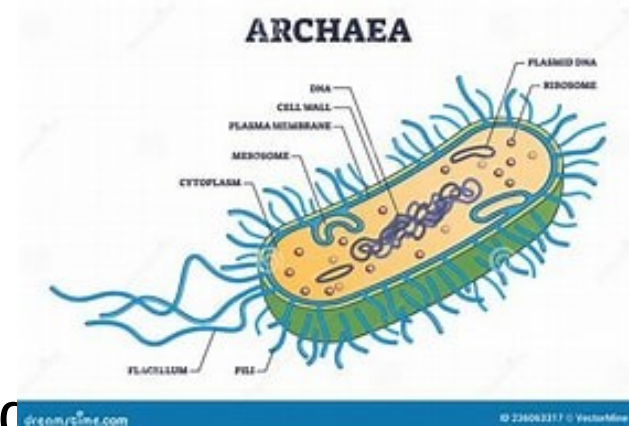
Cell Theory

- All cells share four common components:
 - 1) **Plasma membrane** – outer covering separating cell interiors from its surrounding environment
 - 2) **Cytoplasm** – jelly like liquid within cell in which there are other components
 - 3) **DNA** – the cells genetic material
 - 4) **Ribosomes** – synthesize proteins



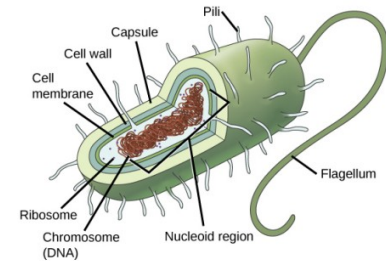
Prokaryotic Cells

- Predominately single celled organisms
 - Bacteria and Archaea
 - Pro = before and kary = nucleus
- **Prokaryote:**
 - Simple
 - Mostly unicellular
 - Lack a nucleus
 - Lack other membrane bound organelles



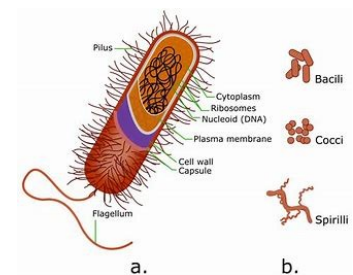
Prokaryotic Cells

- **Prokaryote:**
 - DNA is located in nucleoid (central region)
 - Organized into single circular chromosome
 - Most have a peptidoglycan cell wall
 - Protection and helps maintain shape
 - Many have a polysaccharide capsule
 - Aids in attachment to surfaces
 - Some contain flagella, pili, or fimbriae
 - Flagella – used for locomotion
 - Pili – exchange genetic material during conjugation
 - **Conjugation** – transfer of genetic material through direct contact
 - Fimbriae – bacteria use to attach to host cell



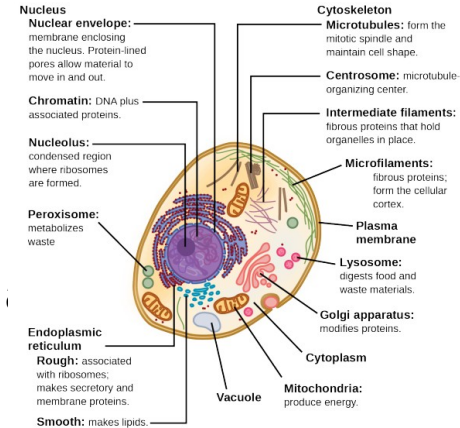
Prokaryotic Cells

- Cell size
 - 0.1 – 5.0 μm
 - Small size allows for quick movement of:
 - Ions and organic molecules into and throughout the cell
 - Waste materials produced in cell out
- Small size is necessary for all cells:
 - If a cell is too large it lacks enough surface area to diffuse fast enough to support the cell
 - As a cell grows it becomes less efficient
 - To fix this a cell can:
 - Divide
 - Increase surface area by folding or changing shape
 - Develop specialized organelles
 - These all lead to more complex cells (eukaryotes)

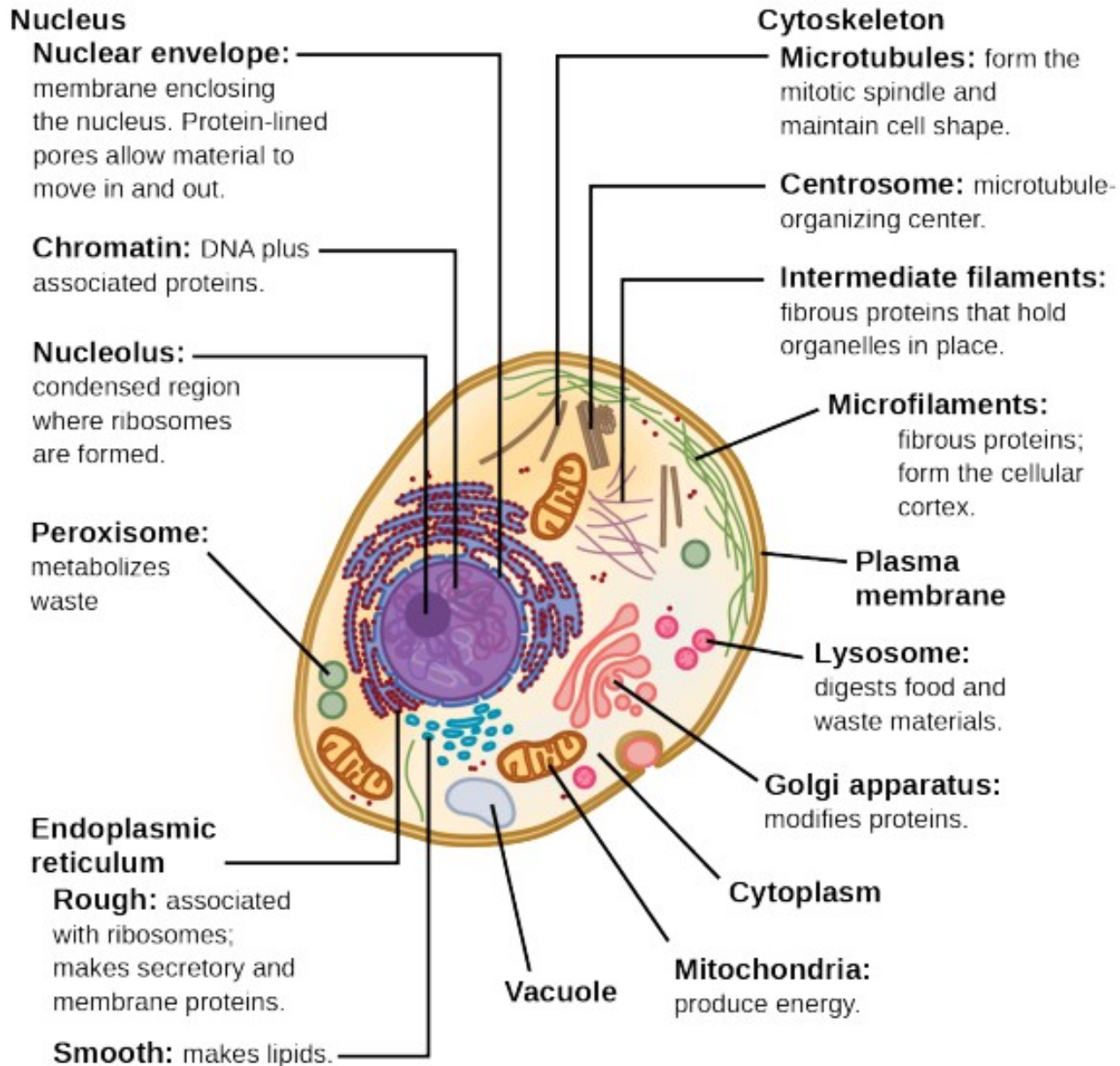


Eukaryotic Cells

- **Eukaryotic cells:**
 - Membrane bound nucleus
 - Numerous membrane bound organelles
 - Organelle = little organ
 - Endoplasmic reticulum
 - Golgi apparatus
 - Chloroplasts
 - Mitochondria
 - Others
 - Allow function to be compartmentalized in:
 - Numerous rod-shaped chromosomes
 - Far more complex than prokaryotic cells



Eukaryotic Cells



Eukaryotic Cells

Plasmodesmata

channels connect
two plant cells

Cell wall maintains
cell shape

**Plasma
membrane**

Cytoplasm

Central Vacuole

filled with cell sap
that maintains
pressure against
cell wall

Cytoskeleton

microtubules
intermediate
filaments
microfilaments

Endoplasmic Reticulum

smooth

rough

Nucleus contains
chromatin, a
nuclear envelope,
and a nucleolus,
as in an animal cell

Ribosomes

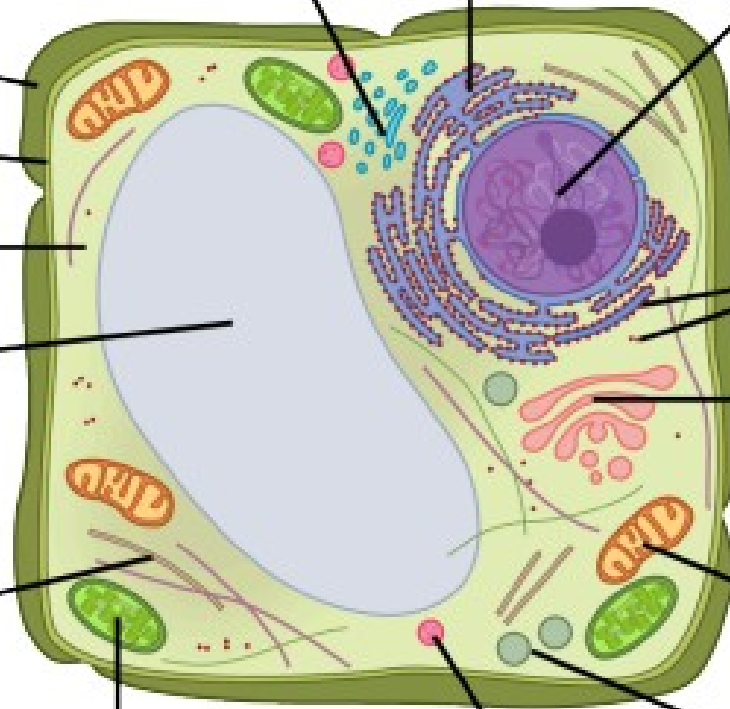
**Golgi
apparatus**

Mitochondria

Peroxisome

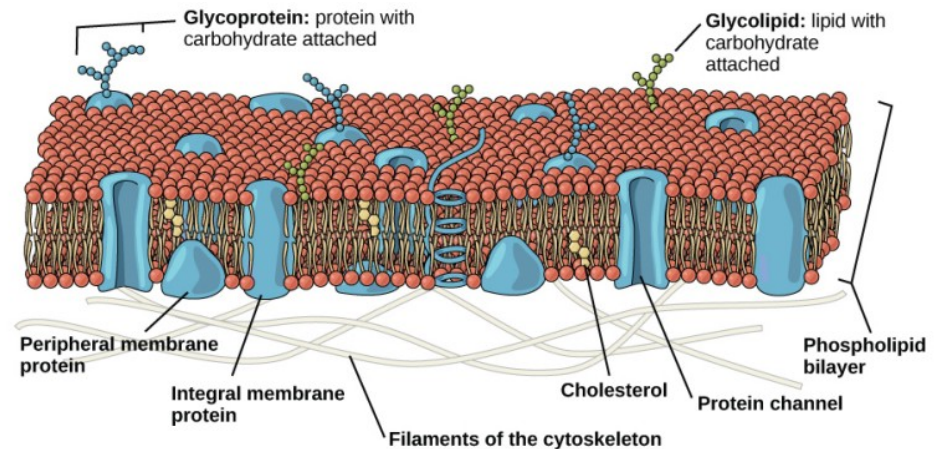
Chloroplast site
of photosynthesis

Plastid store
pigments



Eukaryotic Cells

- **Plasma membrane:**
 - Phospholipid bilayer with embedded proteins
 - Separates cells interiors from environment
 - Controls passage of:
 - Ions
 - Organic molecules
 - Water
 - Oxygen
 - Waste
 - Carbon dioxide

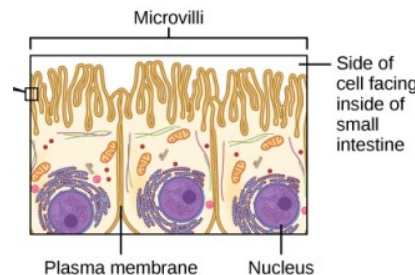


Eukaryotic Cells

- **Plasma membrane:**

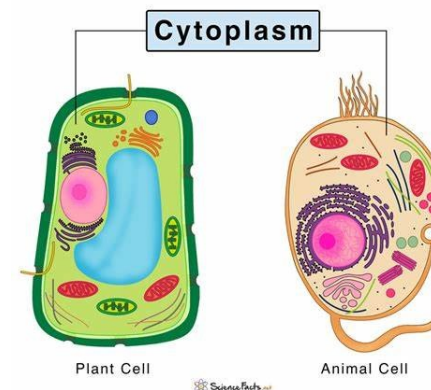
- Real world example:

- Cells that are used for absorption have a plasma membrane that fold into finger like projections
 - Microvilli – finger like folds used in absorption
 - Line the small intestine
 - Absorbs nutrients from our food
 - Celiacs disease
 - Immune response to gluten (protein in wheat, barley, and rye)
 - » Damages the microvilli – prevents absorption
 - » Causes malnutrition, cramping, and diarrhea



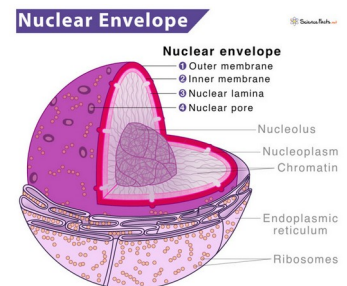
Eukaryotic Cells

- **Cytoplasm** – entire area between plasma membrane and nuclear envelope
 - Consists of a gel like cytosol (cytoskeleton)
 - **Cytosol** – liquid like part of the cytoplasm
 - Is 70-80% water but is a semi solid
 - Due to the proteins contained within it
 - Many other organic molecules and ions as well
 - Many metabolic reactions occur here
 - Including protein synthesis



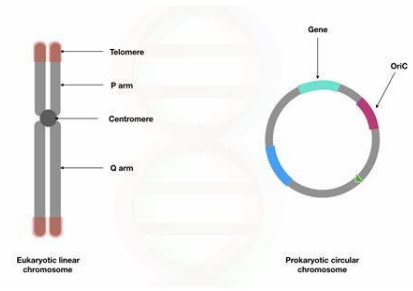
Eukaryotic Cells

- **Nucleus** – houses cells DNA and directs synthesis of ribosomes and proteins
 - Usually most prominent organelle in a cell
- **Nuclear Envelope** – double membrane structure that makes up outermost part of the nucleus
 - Both membranes are phospholipid bilayers
 - Has many pores to control passage between cytoplasm and nucleoplasm
 - Ions, molecules, and RNA
 - **Nucleoplasm** – semi-fluid inside nucleus
 - Where we find chromatin and nucleolus

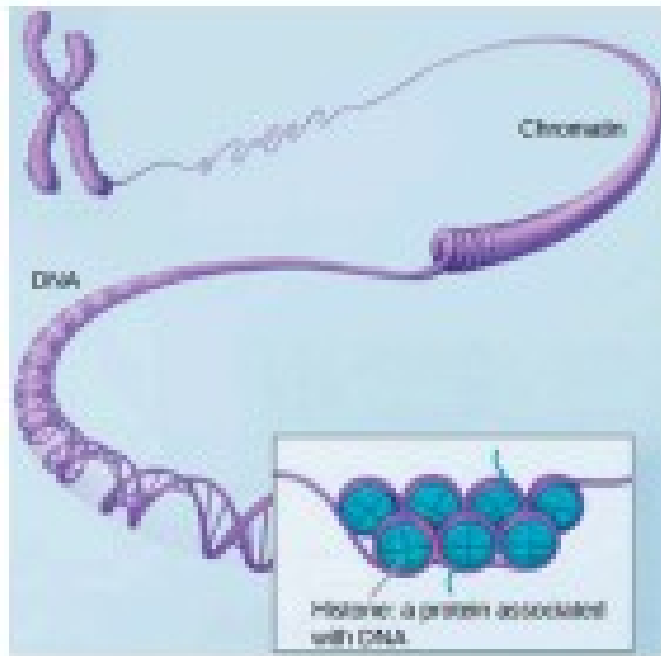


Eukaryotic Cells

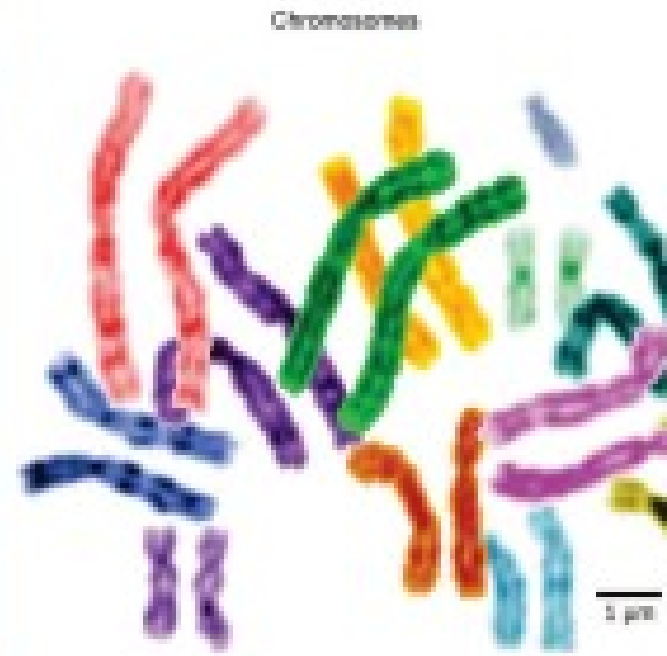
- **Chromosomes** – structures in the nucleus that are made up of DNA
 - Are linear in nature
 - Each eukaryotic organism has a set number in the nucleus of every cell
 - We as humans have 46 chromosomes
 - Fruit flies have 8 chromosomes
 - Only visible when cell is getting ready to divide
 - During most times it is unwound (not condensed)
 - Made of chromatin
 - **Chromatin** – material that make up chromosome



Eukaryotic Cells



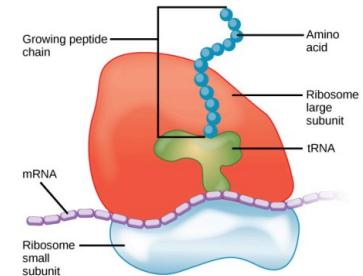
(a)



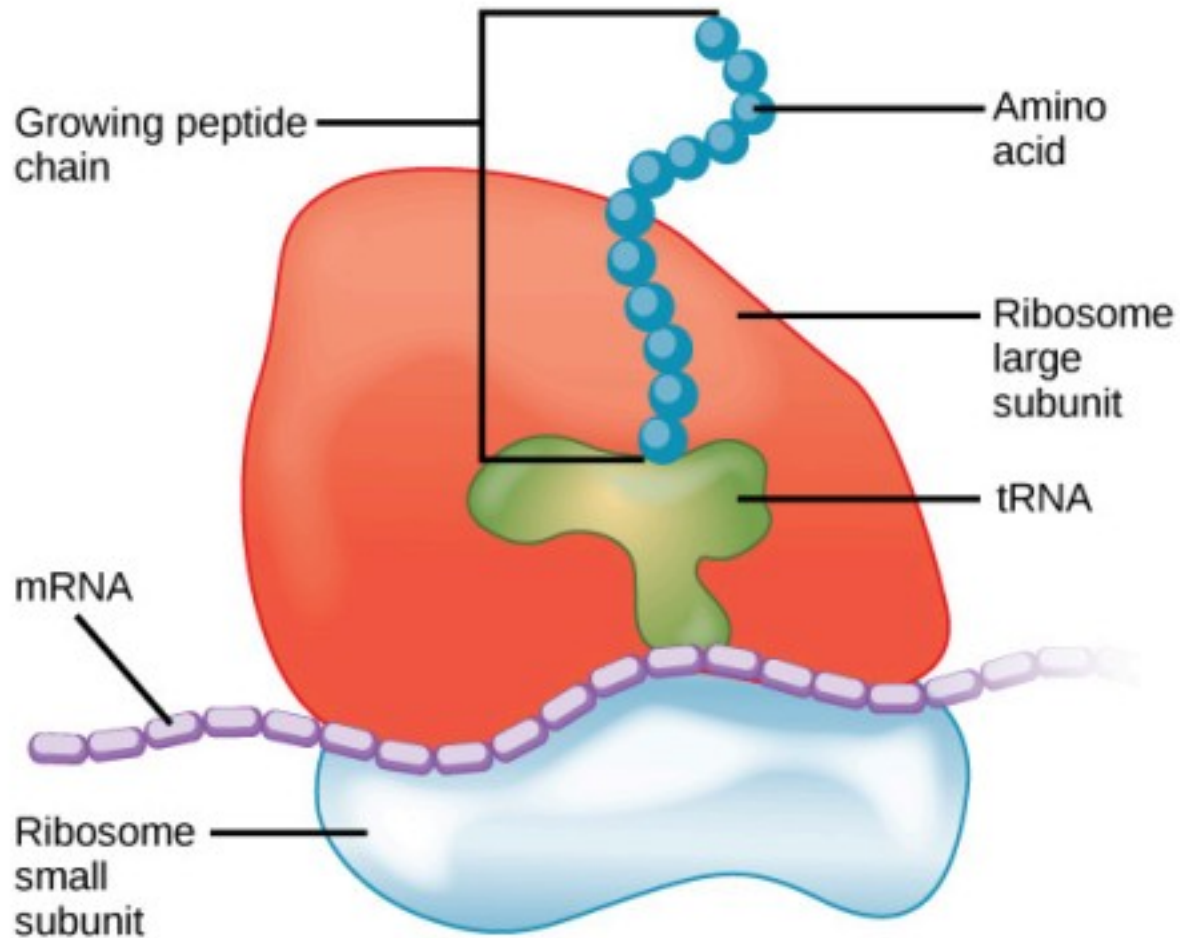
(b)

Eukaryotic Cells

- **Nucleolus** – non-membrane bound organelle in nucleus that synthesizes ribosomal RNA and forms ribosomes
- **Ribosomes** – responsible for protein synthesis
 - Large clusters of RNA and protein
 - Viewed through electron microscope appear as:
 - Single tiny dots floating in cytoplasm
 - **Polyribosomes** – clusters of tiny dots
 - Receive orders from DNA in nucleus
 - DNA is transcribed into mRNA
 - Translates mRNA nitrogenous bases into a specific order
 - Amino acids are the monomers of proteins
 - Occur in almost every cell

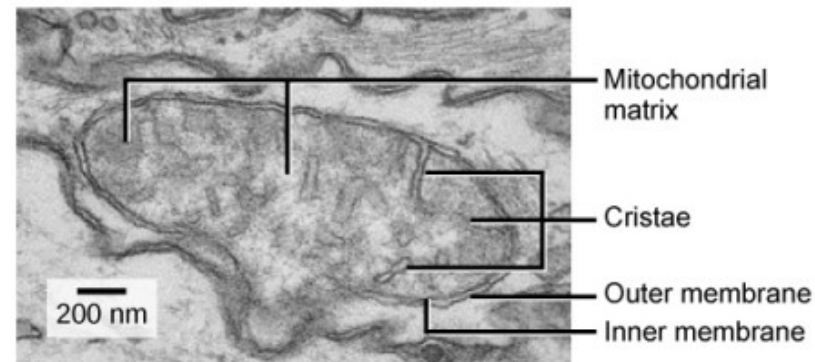


Eukaryotic Cells



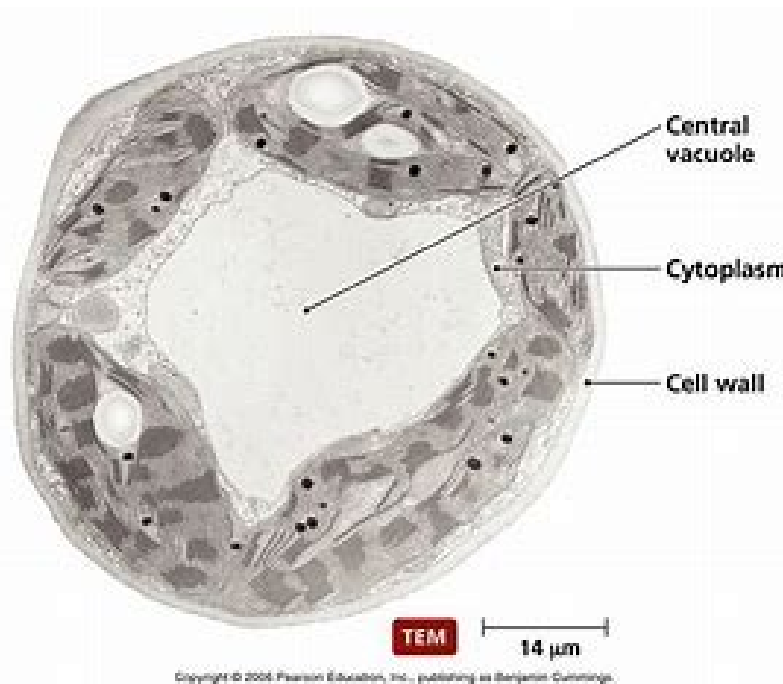
Eukaryotic Cells

- **Mitochondria** – powerhouse of animal and plant cells
 - Make ATP – Adenosine triphosphate
 - Cells main energy carrying molecule
 - Short term stored energy in a cell
 - Produced during cellular respiration
 - Muscle cells have a lot of mitochondria
 - Oval shaped with a double membrane
 - Have own DNA and ribosomes



Eukaryotic Cells

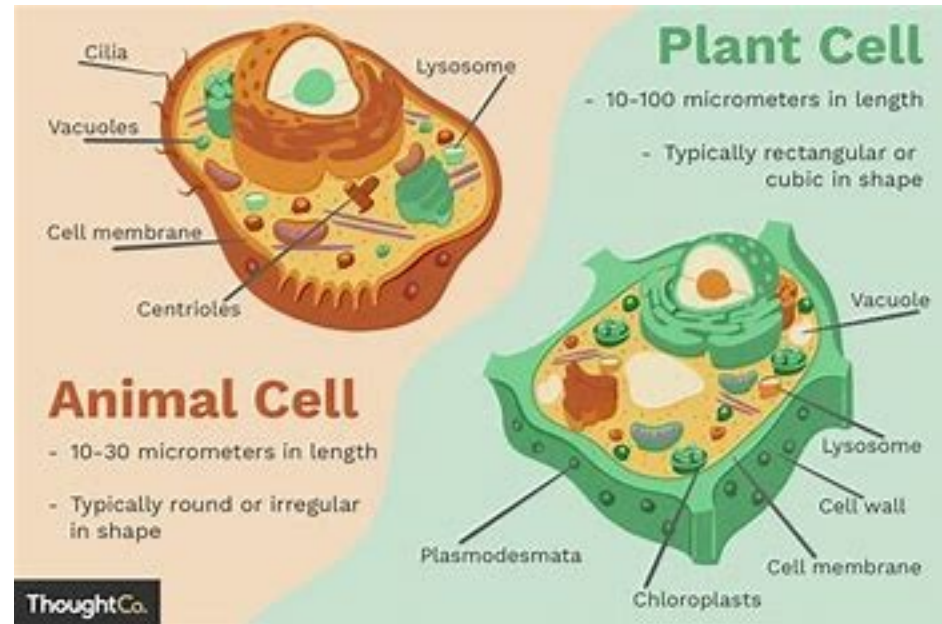
- **Peroxisomes** – break down fatty acids and amino acids and detoxify poisons that enter the body
 - Small, round, only a single membrane
- **Vesicles and Vacuoles** – membrane bound sacs that function in storage and transport



Plant vs. Animal Cells

- **Both Plant and Animal Cells**

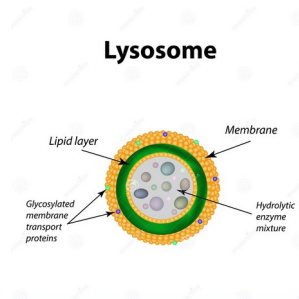
- Plasma membrane
- Cytoplasm
- Nucleus
- Ribosomes
- Mitochondria
- Peroxisomes
- Some have vacuoles
- Endoplasmic reticulum
- Golgi apparatus
- Cytoskeleton



Plant vs. Animal Cells

- **Animal Cells Only**

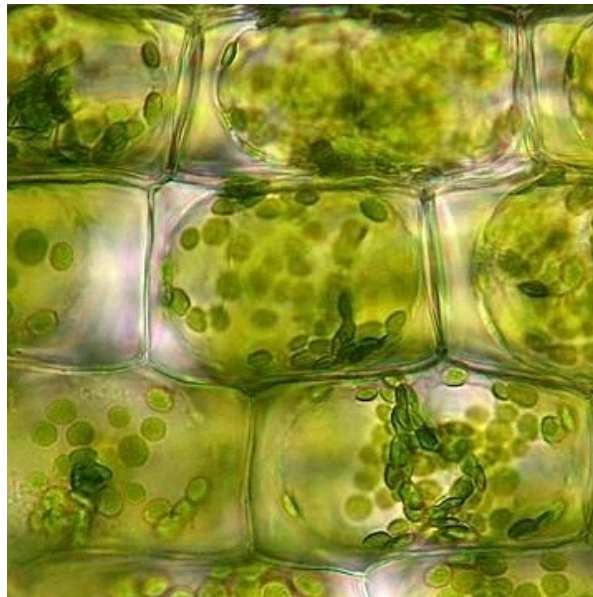
- **Centrosome** – appear to play a role in pulling duplicated chromosomes to opposite sides of the dividing cell
 - Large tube made up of nine sections of three smaller tubes
 - Role is still being studied
 - Cells without centrioles or plant cells still divide
- **Lysosomes** – a cells garbage disposal
 - Contain enzymes that break down:
 - Proteins, polysaccharides, lipids, nucleic acids, old organelles
 - pH in lysosome much lower than cytoplasm
 - Note – in plant cells digestion occurs in the vacuoles



Plant vs. Animal Cells

- **Plant Cells Only**

- **Cell wall** – rigid covering that protects the cell, supplies support, and gives shape to the cell
 - Located outside the plasma membrane
- Some fungi and protists have a cell wall
- Composed of primarily of cellulose (polysaccharide)
 - This is why raw vegetables crunch



Plant vs. Animal Cells

- **Plant Cells Only**

- **Chloroplast** – carry out photosynthesis

- Photosynthesis produces glucose and oxygen

- Plants are Autotrophs

- **Autotroph** – capable of making their own food

- Have their own DNA and ribosomes

- Composed of two membranes

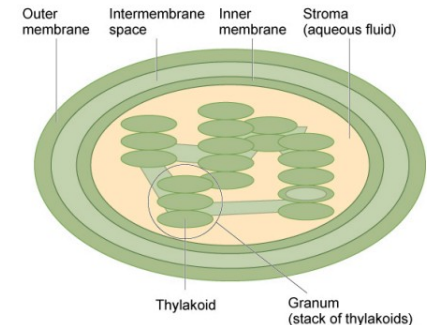
- Inside inner membrane are:

- **Thylakoid** – contains much of the chlorophyll

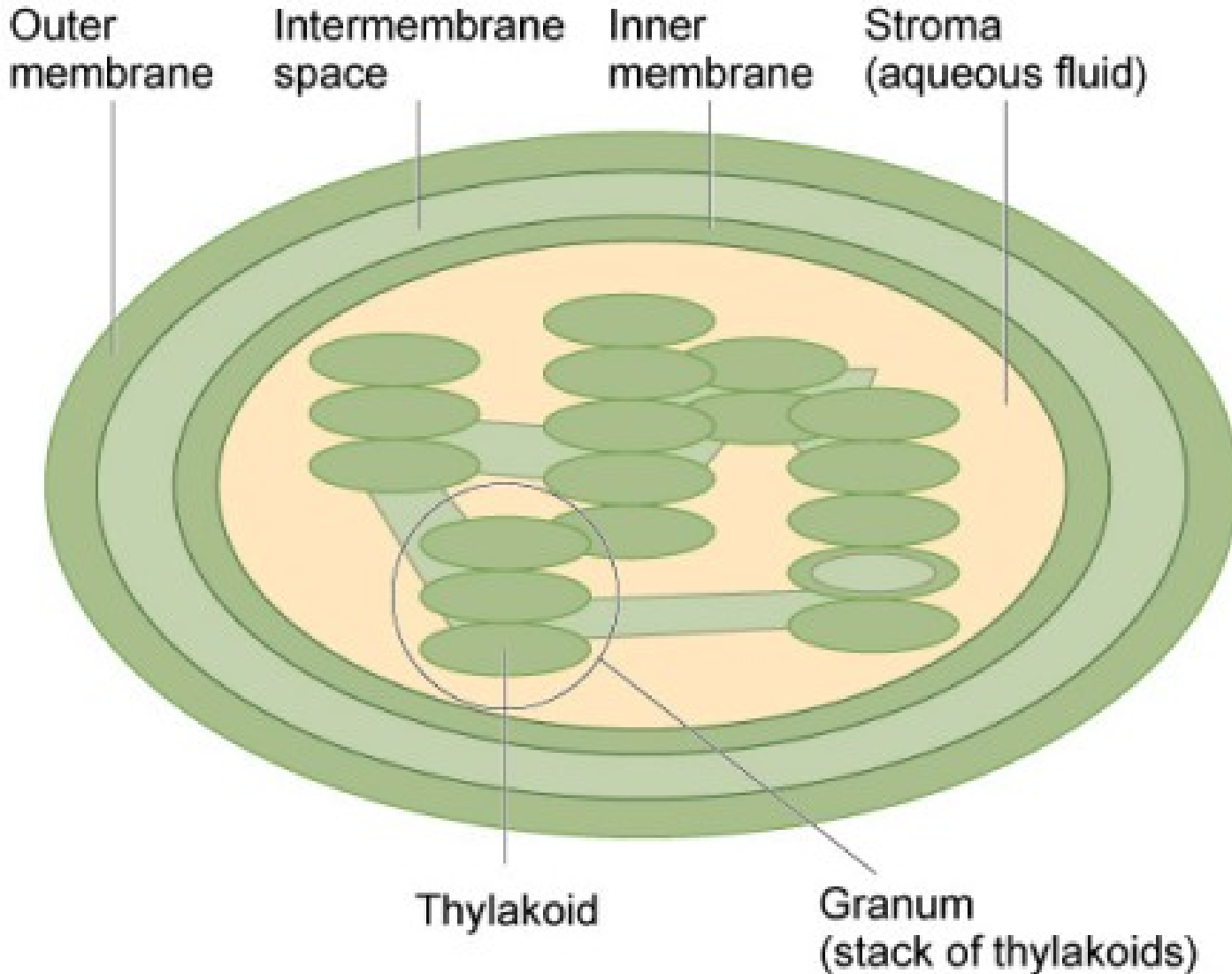
- » **Chlorophyll** – green pigment that captures the light for photosynthesis

- **Granum** – stack of thylakoids

- **Stroma** – liquid contained within the inner membrane



Plant vs. Animal Cells



Plant vs. Animal Cells

- **Plant Cells Only**

- **Central Vacuole** – helps regulate water in a plants changing environment

- Note – animal cells have vacuoles but not a large central vacuole

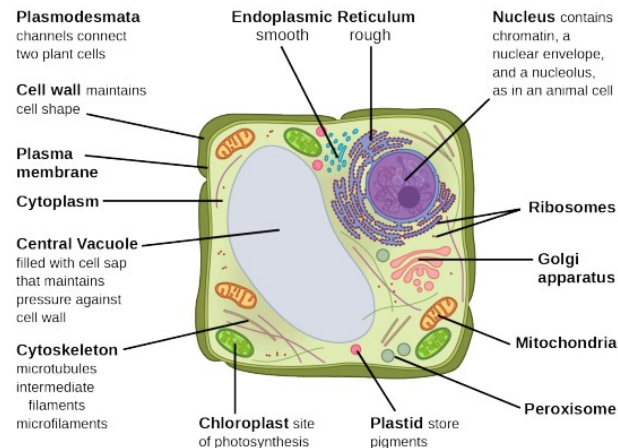
- Wilting of a plant occurs when

- Central vacuole and cytoplasm lose water to dry soil

- Supports cells expansion

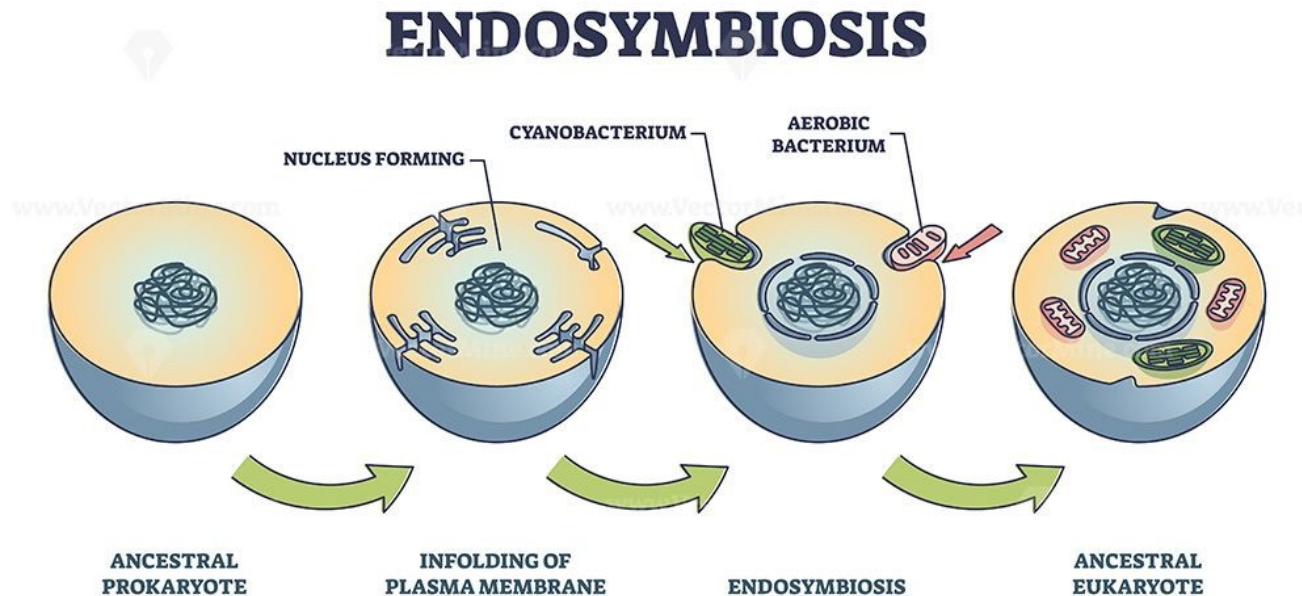
- Central vacuole filled with water cell can expand

- » Does this without making new cytoplasm



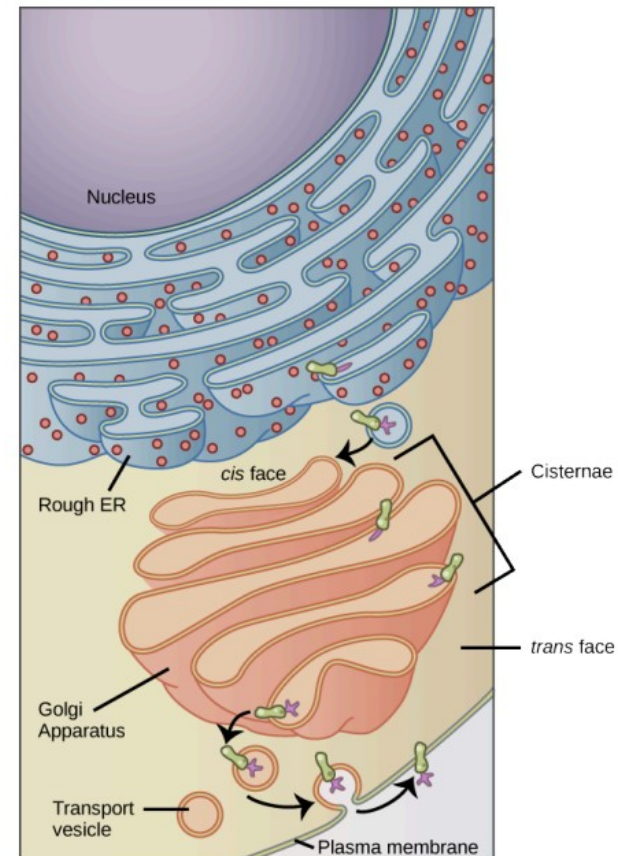
Evolution and Eukaryotic Cells

- Mitochondria and Chloroplasts both have DNA and Ribosomes
 - WHY?
 - Endosymbiosis is best explanation!!



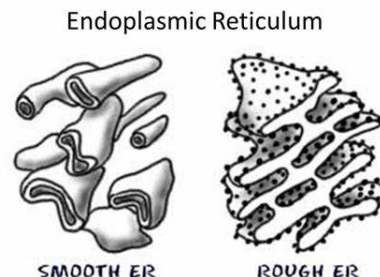
Endomembrane System

- **Endomembrane system** – group of membranes and organelles within the cell that modify, package, and transport lipids and proteins
 - Includes the following:
 - Nuclear envelope
 - Lysosomes
 - Vesicles
 - Endoplasmic reticulum
 - Golgi apparatus



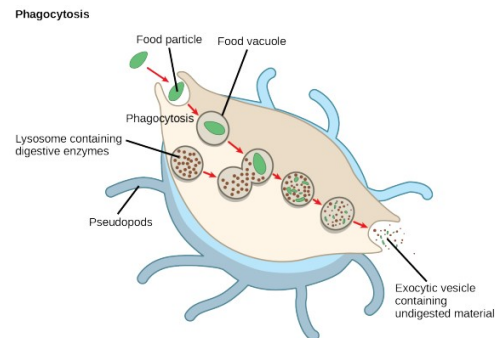
Endomembrane System

- **Endoplasmic reticulum (ER)** – modifies proteins and synthesizes lipids
 - Series of interconnected membranous sacs and tubules
 - The above functions occur in different locations of the ER
 - **Rough ER** – modifies proteins and makes phospholipids
 - Named for the ribosomes attached to its cytoplasmic surface
 - Ribosomes send newly synthesized proteins to Rough ER
 - » Rough ER creates structural modifications (folding or adding side chains)
 - **Smooth ER** – synthesis of carbohydrates, lipids, and steroid hormones, detoxification of medications and poisons, and storing calcium
 - Is a continuation of the rough ER but lack ribosomes or has very few



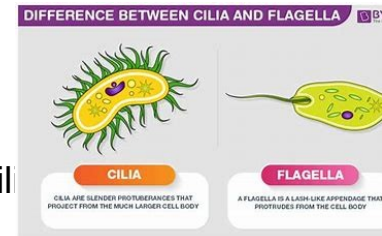
Endomembrane System

- **Golgi Apparatus** – sorting, tagging, packaging, and distributing lipids and proteins from the ER systems
 - Also called the Golgi Body
 - Composed of a series of flattened membrane
 - In plants it synthesizes polysaccharides
 - Incorporated into cell wall
 - Cells that have a large secretory function have lots of Golgi
 - Salivary glands for example
- **Lysosomes** – help to destroy pathogens in animals



Cytoskeleton

- **Cytoskeleton** – maintain the cell's shape, secure some organelles in place, allows cytoplasm and vesicles to move, and enable cells within multicellular organisms to move
 - Series of protein fibers, ions, and organic molecules
 - Three types of fibers:
 - Microfilaments – used in cellular movement
 - Narrowest of all fibers
 - White blood cells – move to where infection is occurring
 - Intermediate filaments – function purely in structure
 - Several strands of fibrous proteins wove together
 - » Keratin strengthens nails and hair
 - Microtubules – help cells resist compression and are part of flagella's, cil



Review Table

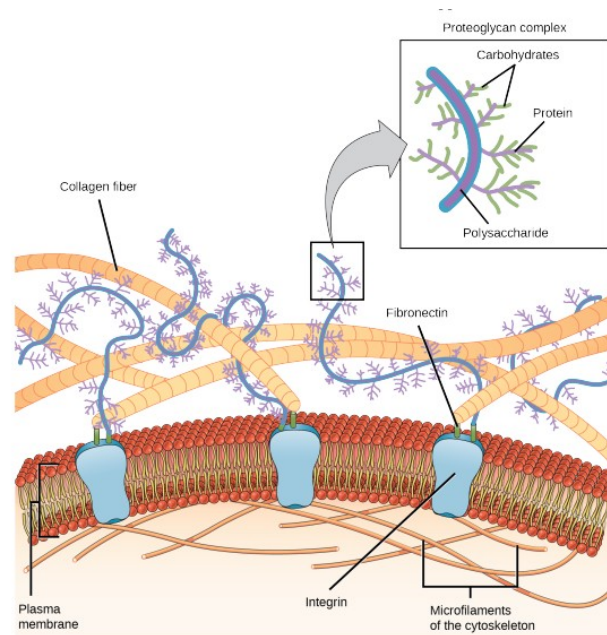
Cell Component	Function	Present in Prokaryotes?	Present in Animal Cells?	Present in Plant Cells?
Plasma membrane	Separates cell from external environment; controls passage of organic molecules, ions, water, oxygen, and wastes into and out of cell	Yes	Yes	Yes
Cytoplasm	Provides turgor pressure to plant cells as fluid inside the central vacuole; site of many metabolic reactions; medium in which organelles are found	Yes	Yes	Yes
Nucleolus	Darkened area within the nucleus where ribosomal subunits are synthesized.	No	Yes	Yes
Nucleus	Cell organelle that houses DNA and directs synthesis of ribosomes and proteins	No	Yes	Yes
Ribosomes	Protein synthesis	Yes	Yes	Yes
Mitochondria	ATP production/cellular respiration	No	Yes	Yes
Peroxisomes	Oxidize and thus break down fatty acids and amino acids, and detoxify poisons	No	Yes	Yes
Vesicles and vacuoles	Storage and transport; digestive function in plant cells	No	Yes	Yes

Review Table

Cell Component	Function	Present in Prokaryotes?	Present in Animal Cells?	Present in Plant Cells?
Centrosome	Unspecified role in cell division in animal cells; microtubule source in animal cells	No	Yes	No
Lysosomes	Digestion of macromolecules; recycling of worn-out organelles	No	Yes	Some
Cell wall	Protection, structural support, and maintenance of cell shape	Yes, primarily peptidoglycan	No	Yes, primarily cellulose
Chloroplasts	Photosynthesis	No	No	Yes
Endoplasmic reticulum	Modifies proteins and synthesizes lipids	No	Yes	Yes
Golgi apparatus	Modifies, sorts, tags, packages, and distributes lipids and proteins	No	Yes	Yes
Cytoskeleton	Maintains cell's shape, secures organelles in specific positions, allows cytoplasm and vesicles to move within cell, and enables unicellular organisms to move independently	Yes	Yes	Yes
Flagella	Cellular locomotion	Some	Some	No, except for some plant sperm cells
Cilia	Cellular locomotion, movement of particles along plasma membrane's extracellular surface, and filtration	Some	Some	No

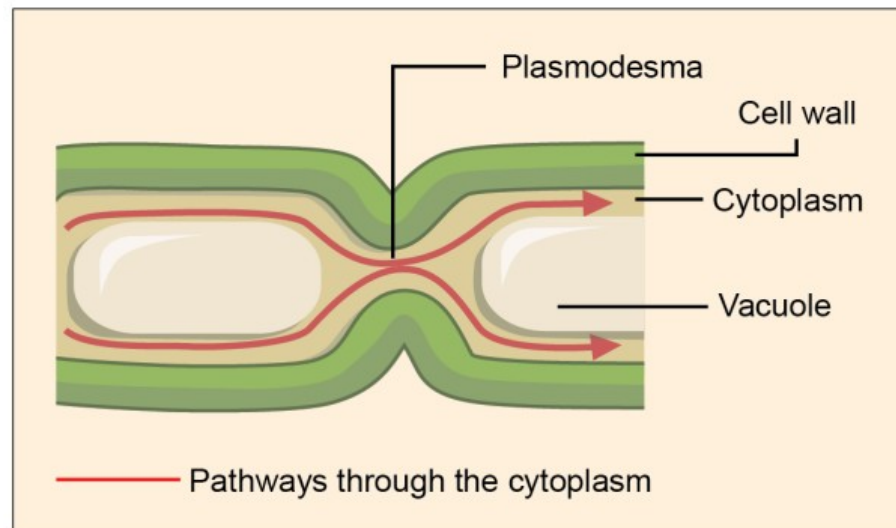
Extra Cellular Matrix

- **Extracellular matrix** – holds cells together to form tissues allows cells in tissue to communicate
 - Made of materials released from cells into the intercellular space
 - Proteins and carbohydrates containing proteins
 - Blood clotting is an example



Intercellular Junctions

- Cells can communicate via:
 - Extracellular matrix
 - Direct contact
 - Intercellular junctions
- **Plasmodesmata** – junctions between plant cells
 - Channels between plant cells that connect cytoplasm
 - Important in xylem and phloem transport



Intercellular Junctions

- Animal cells
 - **Tight junction** – watertight seal that prevents materials from leaking between two animal cells
 - Bladder of skin cells
 - **Desmosomes** – act like a spot weld between animal cells maintain the sheet-like formation
 - Found in organs and tissues that stretch
 - Heart, skin, and muscles
 - **Gap junctions** – cells with a channel between them
 - Similar to plasmodesmata in plant cells
 - Allow transport between two cells
 - Example = cardiac muscle – allows electrical signal to contract to pass through quickly so all contract at the same time

Next Time

- Structure and Function of Plasma Membranes (Chapter 5)

